

Muhammad Moosa Bin Naseem

SOFTWARE ENGINEER · DATA SCIENTIST

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Skills

Deep Learning Frameworks

Generative AI and LLMs

Programming and High-Performance Computing

MLOps and Quantization

Back-end and Networking

Databases and Vector DBs

Pytorch, TensorFlow

HuggingFace Transformers, LangChain, Agents, MCP

Python, C++, CUDA, Rust, Go

Docker, Llama.cpp, Ollama, ONNX, MLFlow

Flask, Tokio, TCP/UDP, REST

MongoDB, PostgreSQL, Clickhouse, Qdrant, Pinecone

Work Experience

Virtual Remittance Gateway (VRG) — AI Research and Development, GRC

Karachi, Pakistan

DATA SCIENTIST

December 2024 — Present

- Building software solutions to optimize business workflows and in-house tools for management and analytics
- Built a full stack analytics engine serving different dashboards, REST APIs and an LLM Chat Assistant ; with React, Node.JS/Express, Python and Oracle.
 - Utilized **Python+Oracle** data pipeline to join transaction tables of >30M and >14M rows into an aggregated json snapshot reducing load time from **≈60 seconds to < 1 second** .
 - Built an Agentic Chat Assistant using LLM tool calling with **Ollama**; wrote 6 different tools for data-retrieval, with a multi-tool call loop, thread-based session persistence, and **SQLite DB** for chat history storage and restarts.
 - Implemented layered guard-rails for AI security against hallucination and prompt injection: a Pre-flight LLM classifier to detect the scope of queries, and hardened system prompt boundaries to ensure tool-returned answers.
 - Built a client side geospatial Agent-Locator utility using **Deck.gl**, **MapLibreGL** and **SuperCluster** to track activity of AIO Agents across Pakistan.
 - Built a zero-dependency Node gateway serving all SPAs and consolidated all four different services onto a single port.
- Relying on AI coding tools **ClaudeCode**, **OpenCode**, **Codex** for rapid prototyping, exploring new technologies and creating secure-by-design applications.
- Built an internal suite utilizing **Python/Flask framework** to automate daily reporting for State Bank of Pakistan and partner banks for regulatory compliance.
- Working closely with Cybersecurity team to perform vulnerability assessment of in-house apps with **ZAP**

Askari Technologies — Software Engineering Team

Karachi, Pakistan

JUNIOR AI DEVELOPER

August 2024 — September 2024

- Built multiple AI and software solutions for different international clients
- Built and Deployed a RAG pipeline for **Telecof**, for FAQs and appointment booking via WhatsApp. Deployed using **Digital Ocean** and **Twilio**
- Built high-performance APIs to transmit trading signals between different platforms e.g. **SierraCharts** and **MetaTrader5** with millisecond precision
- Wrote backend services and fine-tuned transformers using **C++**, **Python**, **PyDantic**, **Huggingface Transformers**
- Built an application to generate Arabic poetry for an educational institution, deployed using **IBM Watson**
- Built a bot to scrape and share trending videos from Youtube and Facebook using **Youtube Data API** and **GraphAPI**
- Worked on QR code generation and obfuscation with AI generated images

Gerry's — Software Engineering Team

Karachi, Pakistan

WEB DEVELOPMENT INTERN

Fall 2023

- Wireframed and Developed the frontend for Gerry's official website
- Worked on developing the frontend for a Gerry's group subsidiary

JS Bank Ltd. — Ancillary Apps Development

Karachi, Pakistan

DATA ENGINEERING INTERN

Summer 2016

- Wrote **SQL Queries** for Ad Hoc data extraction and manipulation
- Worked on **Microsoft SSIS and SSMS** to perform ETL processes and aggregations

Projects

Hades: AI Based intrusion detection system

(Paper)

UNDERGRADUATE RESEARCHER

- Worked on core model training using LSTM-AE for feature compression and Catboost algorithm for classification
- Fine tuned and compared the hybrid learning technique with different machine learning algorithms
- Wrote backend services to transmit flagged IPs over to the dashboard

Pherrous: Distributed Network Computing Infrastructure

(Link)

SOFTWARE ENGINEER

- Utilizing Rust to create a network computing system inspired by BOINC as a practice project
- Defined a network communication protocol to transmit files
- Currently working on a module to track computation progress and communicate results back to the host
- Utilizing Ratatui framework for user friendly TUI for the client and the server

Forex Trend predictor

DATA SCIENTIST

- Built a Forex prices and trend predictor using Bi-Directional LSTMs, achieving high accuracy over short term predictions
- Built the Neural Network using **PyTorch**, collected data on demand using **yFinance** and an interactive frontend with **Tkinter**

DOOM AI Agent

DATA SCIENTIST

- Trained an Agent via Reinforcement Learning to play the iconic DOOM 1993
- Utilized **Stable Baselines 3** to train the agent using **PPO** algorithm
- Used **VizDOOM** for environment models and **TensorBoard** for Visualization

Education

BS in Computer Science

Karachi, Pakistan

INSTITUTE OF BUSINESS MANAGEMENT, KARACHI

2020 — 2024

- **Relevant Coursework**— Machine Learning, Data Science and Analytics, Software Engineering, Computer Security, Programming Language Theory, Compiler Theory, Operating Systems and Networking, Cryptography, Networked and Distributed Systems
- **Thesis**— AI based Intrusion Detection System using Hybrid Learning